LAPTOP RECOMMENDATION SYSTEM

An intelligent laptop recommendation system developed using Fuzzy Logic (Mamdani Type-1) and a custom-built dataset to provide personalized laptop suggestions based on user preferences and technical requirements.

Project Overview

This project helps users choose suitable laptops by analysing inputs such as:

- Budget
- RAM
- Processor
- Storage
- Usage Requirements

The recommendation process is powered by a Mamdani Type-1 Fuzzy Inference System (FIS) created in MATLAB, which evaluates user preferences and generates accurate laptop recommendations.

The system also provides direct shopping links to Amazon and Flipkart, allowing users to instantly explore and purchase the recommended laptops from trusted e-commerce platforms.

The web application is built using:

- Python (Flask)
- HTML
- CSS
- JSON Dataset

Technologies Used

- MATLAB (Fuzzy Logic Toolbox)
- Mamdani Type-1 Fuzzy Inference System
- Python Flask
- HTML5
- CSS3
- JSON

Project Workflow

1. Created a custom laptop dataset.
2. Designed a Mamdani Type-1 Fuzzy Logic model in MATLAB.
3. Generated fuzzy rules and recommendation logic.

4. Exported the FIS model.
5. Developed a Flask-based web application.
6. Built frontend pages using HTML and CSS.
7. Integrated recommendation logic with the user interface.
8. Added direct redirection to Amazon and Flipkart product pages.

Project Structure

Laptop-Recommendation-System/

static/
style.css
templates/
index.html
landing.html
LaptopRecommender.fis
laptops.json
app.py

Features

- Personalized laptop recommendations
- Fuzzy logic–based intelligent decision making
- User-friendly interface
- Custom dataset integration
- Budget and specification analysis
- Direct redirection to Amazon and Flipkart shopping pages
- Fast and lightweight Flask web application
- Smart recommendation system based on user preferences

Specialities of the Project

- Uses **Mamdani Type-1 Fuzzy Logic** for intelligent recommendations
- Developed using a **self-created dataset**
- Helps users avoid manual laptop comparison
- Provides quick access to online purchasing platforms
- Simplifies laptop selection for students, professionals, and gamers

- Combines AI concepts with web development for practical use

Future Enhancements

- AI/ML-based advanced recommendation engine
- Real-time price tracking
- User authentication system
- Advanced filtering and comparison
- Database integration
- Recommendation accuracy improvements

Authors

Developed as an intelligent recommendation system using Fuzzy Logic, Web Technologies, and a custom-built laptop dataset.

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